

Human-AI Collaboration in Scientific Writing and Research Workflows

Abstract

Human-AI collaboration in scientific work has moved rapidly from narrow assistance tools for grammar checking, citation recommendation, and document retrieval to increasingly integrated systems that participate in ideation, literature review, coding, analysis, drafting, revision, peer review, and even end-to-end agentic research pipelines. This survey synthesizes the literature on how such systems are designed, used, and evaluated across the scientific workflow. It argues that the field is best understood not as a single question of “whether AI can write science,” but as a set of collaboration design problems concerning task decomposition, epistemic authority, provenance, verification burden, writer agency, and institutional accountability. Drawing on research from human-computer interaction, natural language processing, information retrieval, digital libraries, science of science, and agentic AI, the survey organizes prior work into method families: literature discovery and synthesis; ideation and hypothesis generation; coding, analysis, and experimental support; manuscript planning, drafting, revision, and citation grounding; review and publication support; and orchestration interfaces for mixed-initiative workflows. Across these areas, the empirical record suggests three recurrent patterns. First, AI can reduce friction and expand search breadth, but often shifts work rather than removing it, creating verification, curation, and integration overhead. Second, the most successful systems are not fully automatic authorship engines but interfaces that preserve human control over intermediate artifacts, revisions, evidence, and final decisions. Third, the central technical bottlenecks are no longer only fluency and coherence, but citation faithfulness, source attribution, reproducibility, authorship transparency, and workflow-level evaluation. The survey concludes that progress will depend on moving from isolated text generation benchmarks to collaboration-centered research infrastructures that measure how people and AI systems jointly produce reliable, attributable, and pedagogically valuable scientific outcomes. ¹

Introduction

Human-AI collaboration has become a central organizing idea in contemporary AI research because many consequential tasks require combining machine speed, retrieval breadth, and pattern induction with human judgment, responsibility, and domain understanding [1–10]. Scientific writing and research workflows are now one of the most consequential instantiations of this broader agenda. In these settings, researchers do not merely ask models to “generate text.” They use AI systems to search literature, cluster themes, summarize papers, suggest hypotheses, write code, interpret results, draft sections, revise prose, format citations, simulate review, and visualize contributions [37–60]. Large-scale evidence suggests that this transformation is no longer anecdotal: one survey of 816 verified research authors reported that a large majority had already integrated LLMs into at least part of their workflow, while corpus-level analyses of preprints and journal articles documented a measurable and rising share of LLM-modified scientific text, with especially rapid growth in computer science [41–44]. ²

The rapid uptake of AI in scientific writing has created a conceptual mismatch between public debate and research reality. Public debate often focuses on the binary question of whether AI “writes papers.” The literature instead shows a spectrum of collaboration regimes. Some systems provide bounded assistance, such as inline critique, citation recommendation, or argument scaffolding [11–18, 30–31, 37–40]. Others

support mixed-initiative authoring in which a human researcher repeatedly steers ideation, accepts or rejects revisions, rewrites AI text, and validates sources [14, 16–17, 19–29, 59–60]. At the far end are agentic pipelines that attempt to automate large portions of the scientific cycle, including idea generation, coding, experimentation, paper drafting, and review [52–60, 77–92, 106–116]. The crucial research question is therefore not simply capability, but allocation: which parts of scientific work can be delegated, which must remain under human control, and how should interfaces expose provenance, uncertainty, and authorship? ³

Scientific writing is a particularly revealing domain for studying collaboration because writing is both a final product and a thinking process. Prior work on writing support systems emphasizes that writers use tools not only to correct language but to externalize thought, sustain motivation, test ideas, structure argument, preserve voice, and negotiate ownership [11–13, 17–29, 141–142]. Recent meta-analyses of writing support tools show that the design space has expanded far beyond grammar and spelling toward systems that intervene across ideation, drafting, revision, and reflection, yet many writer needs remain insufficiently served, especially those tied to authenticity, higher-order reasoning, and long-horizon organization [141–142]. The scientific case sharpens these concerns further, since text must be evidentially grounded, citationally faithful, methodologically precise, and socially accountable. A system that is helpful for creative writing may still be inadequate for literature review, methods reporting, or peer review. ⁴

A second reason the area matters is that scientific workflows contain multiple epistemically distinct subtasks. Literature retrieval demands recall, ranking, and corpus coverage [61–76, 131–136]; ideation demands novelty while remaining plausible and testable [77–92]; coding and data analysis require executable correctness and domain-sensitive verification [93–105]; drafting and revision require discourse planning, style adaptation, and citation grounding [37–40, 45–46, 68–76]; and reviewing requires critique, calibration, and protection against gaming [106–116]. This heterogeneity helps explain why simplistic claims about complete scientific automation are premature. Models may appear strong on fluent summarization while remaining weak on citation attribution or reproducibility. Indeed, recent benchmarks suggest a large gap between human experts and frontier models in open-ended scientific claim attribution, even when agents are permitted to search and read papers [74–76, 136]. ⁵

This survey takes a collaboration-centered view of the field. Rather than reviewing only “AI for science” systems or only “AI writing” systems, it examines the overlapping literature on how researchers and AI systems jointly produce scientific artifacts. The result is a taxonomy organized around workflow stages and collaboration mechanisms, with particular attention to evidence grounding, interaction design, evaluation methodology, and research integrity. The survey’s main claim is that the most important technical frontier in this area is no longer raw text generation quality, but the design of trustworthy, inspectable, and institutionally compatible collaborations between people and models. ⁶

Methods

Scope and survey protocol

This survey targets scholarly work on human-AI collaboration in scientific writing and research workflows. The scope includes peer-reviewed conference papers, journal articles, academic workshop papers, and arXiv preprints that study one or more of the following: collaborative writing systems, scientific writing assistance, literature review support, hypothesis generation, coding or data-analysis assistance for researchers, automated or AI-assisted peer review, and workflow orchestration for research tasks [11–145]. The survey deliberately excludes purely commercial product descriptions, general media commentary, and non-scholarly web sources. It also avoids treating all “AI for science” work as

relevant. The focus is specifically on systems and studies that engage with collaboration around scientific reasoning, research artifacts, or scholarly communication, not only domain prediction performance in fields such as chemistry or biology. ⁷

The literature base is heterogeneous. Some papers propose systems and datasets [14, 37–40, 61–76, 93–116]. Others report user studies, field studies, or corpus analyses of real researchers and writers [14, 16–18, 21–35, 41–46, 93, 100–105, 117–130]. Still others offer surveys, conceptual analyses, or position papers that synthesize emerging concerns about authorship, trust, and human-centered design [10, 47–57, 66–68, 80, 90, 107, 110, 117–127, 141–145]. Because the field is changing quickly, this survey privileges papers that make direct empirical or methodological contributions to collaboration, while using broader survey papers to situate trends and terminology. Recent surveys on LLMs in scientific research, scientific discovery, peer review, and human-AI collaboration help confirm that this field has diversified into several subareas rather than a single research stream [47–57, 80, 90, 107, 110]. ⁸

Collaboration models across the research lifecycle

The surveyed literature can be organized by both workflow stage and collaboration regime. Along the workflow axis, systems support discovery, reading, synthesis, ideation, analysis, drafting, revision, review, and publication. Along the collaboration axis, they range from assistive to mixed-initiative to quasi-autonomous. Assistive systems provide bounded interventions such as citation suggestions, stylistic revisions, critique prompts, or workflow scaffolds [30–31, 37–40, 68–76, 106]. Mixed-initiative systems allow turn-taking, iterative editing, selective acceptance, or feedback-driven refinement [14, 16–18, 19–29, 59–67, 93–99]. Quasi-autonomous systems attempt to execute longer research trajectories with tool use, memory, planning, and internal review [52–60, 77–89, 106–116]. ⁹

A useful synthesis from the broader human-AI collaboration literature is that collaboration quality depends not only on model accuracy but on whether the interface enables calibrated delegation, efficient error detection, and appropriate accountability [1–10]. In scientific workflows, this means exposing sources, preserving revision histories, enabling comparison among evidence candidates, and allowing the human researcher to retain epistemic control over claims and conclusions. The growing turn toward “controllable” academic revision, human-centered writing tutors, draft transparency tools, and workflow provenance systems reflects this design shift [21, 27–29, 39–40, 59–60, 97]. ¹⁰

The table below summarizes the collaboration-oriented method families that recur across the literature.

Workflow family	Typical human role	Typical AI role	Representative references	Primary evaluation focus
Literature discovery and synthesis	Define topic, inspect papers, verify evidence	Retrieve, cluster, summarize, compare	[61–76, 131–136]	relevance, coverage, faithfulness, verification burden
Ideation and hypothesis generation	Frame problem, judge novelty and feasibility	Generate candidates, expand search, connect literatures	[77–92]	novelty, plausibility, usefulness, downstream utility

Workflow family	Typical human role	Typical AI role	Representative references	Primary evaluation focus
Coding, analysis, experimentation	Specify goals, test outputs, interpret results	Generate code, wrangle data, propose analyses	[93–105]	correctness, efficiency, debugging burden, interpretability
Drafting and revision	Plan argument, rewrite, approve claims	Outline, draft, critique, revise prose	[37–40, 45–46, 68–76]	text quality, citation fidelity, controllability, ownership
Review and publication support	Critique, decide, revise, disclose use	Generate reviews, meta-reviews, draft responses	[106–116]	alignment, constructiveness, gaming resistance
Orchestrated workflow systems	Supervise pipeline, intervene at key steps	Coordinate multi-agent subtasks with tools	[52–60, 77–89, 97, 115–116]	end-to-end utility, traceability, reproducibility

Literature discovery and synthesis

The literature review stage has become one of the most active areas of collaboration research because it is simultaneously labor-intensive, high-value, and structurally well suited to retrieval-augmented generation. Early work emphasized citation recommendation and scientific summarization [131–134]. More recent systems move beyond recommending single papers toward full workflows for related-work authoring, including search, paper selection, note comparison, thematic grouping, citation text generation, and draft synthesis [61–76]. This trajectory reflects an important conceptual change: the task is no longer treated as “generate a related-work paragraph,” but as a sequence of interpretive and evidential decisions in which retrieval, reading, and writing are tightly coupled [66–73]. ¹¹

A persistent theme in this literature is the move from isolated generation toward verifiable reading support. Systems such as LitLLM, AI Literature Review Suite, and vitalITy 2 use RAG pipelines to retrieve papers and generate summaries or thematic syntheses [61–63]. Studies of actual researchers using LLMs for literature review report broad enthusiasm for acceleration, but identify three recurring pain points: lack of trust in outputs, high verification burden, and fragmentation across tools [64–67]. Design responses include structured views of literature, stable citation grounding, side-by-side comparison, user preference preservation, and opportunities for incremental feedback rather than one-shot synthesis [66–67]. This is one of the clearest examples where collaboration research has moved from “can the model summarize?” to “how do we reduce the cost of checking it?” ¹²

Related-work generation offers an especially revealing subproblem. Citation text generation and related-work generation have a pre-LLM history in NLP and IR [68–73, 131–134], but recent work rethinks the task around structure, novelty statements, citation ordering, and multi-agent pipelines [69–73]. The best systems increasingly decompose the problem into selecting papers, reading them, constructing a faceted plan, and then drafting text, rather than directly generating a polished section. This decomposition matters because literature review is not merely compressive summarization; it is an argumentative act that positions a new contribution relative to prior work. Papers on structured related-work generation

therefore foreground paragraph structure, topical segmentation, and explicit novelty framing [69–73].

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Yet the hardest bottleneck is not prose quality but source attribution. Citation attribution benchmarks show that open-ended scientific citation remains challenging even for strong models and tool-augmented agents [74–76, 136]. This finding has two major implications. First, systems that produce plausible literature reviews without grounded citation mechanisms should not be treated as reliable scholarly coauthors. Second, faithful literature review support likely requires tighter integration of search, retrieval, reading traces, claim-level provenance, and uncertainty communication than current chat-style interfaces provide [66–67, 74–76]. 5

Ideation, hypothesis generation, and planning

Idea generation is frequently promoted as the most exciting possibility of AI-assisted science, but the literature suggests a more nuanced picture. On the positive side, LLMs can search large literatures, recombine concepts across domains, propose alternative framings, and broaden the candidate idea space [77–90]. ResearchAgent formalized iterative research idea generation over scientific literature as a multi-step process [77]. SciMuse and related systems incorporated knowledge graphs and large-scale paper corpora to generate and rank ideas, including evaluations by large cohorts of research group leaders [79]. Surveys on hypothesis generation and scientific ideation now treat novelty, plausibility, feasibility, and testability as distinct objectives rather than one overall quality dimension [80, 83, 90–91].

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The central collaboration problem in AI-assisted ideation is that idea quality is multi-objective and domain relative. A model may generate a novel but infeasible proposal, or a plausible but uninteresting variant of existing work. This motivates research on grounded ideation, data-aware idea generation, benchmark design, and Bayesian approaches to open-ended discovery [81–85, 88–92]. Human researchers are essential here not merely to filter outputs, but to judge latent criteria that datasets capture poorly: whether an idea is worth pursuing, whether evidence requirements are reasonable, whether the proposed framing fits institutional constraints, and whether novelty is genuine rather than stylistic. The emerging benchmark literature therefore measures more than lexical diversity; it asks whether ideas are useful for actual research planning [82, 88, 91–92]. 15

Agentic “AI scientist” systems push ideation into end-to-end automation [52–55, 86–89]. These systems can propose ideas, write code, run experiments, generate figures, draft papers, and simulate review. They are important because they reveal both the ambition and the fragility of current paradigms. The strongest lesson from this literature is that end-to-end automation is more compelling as a stress test of workflow decomposition than as evidence that humans can be removed from scientific work. Position papers and surveys increasingly argue that these systems function better as co-scientists or workflow accelerators than as autonomous researchers, because evaluation, problem selection, evidential standards, and societal accountability remain deeply human concerns [53, 55–57]. 16

Coding, analysis, and experimental support

Scientific research increasingly depends on programming, scripting, and data analysis, making code-generating models highly relevant to research workflows [93–105]. Interview and log analyses show that scientists often use LLMs not only as code generators but as information retrieval systems for unfamiliar languages, libraries, and analytic patterns [93–94]. This use is doubly significant. It expands access for researchers with uneven programming backgrounds, but it also risks silently shaping analytic decisions through plausible yet unverified code templates. The collaboration question is therefore not simply

whether models can write code, but whether interfaces help researchers inspect assumptions, validate computations, and maintain reproducible analytical intent [93–99]. ¹⁷

Mixed-initiative data-analysis systems make this issue explicit. Work on interaction modes, user agency, and mixed-initiative data wrangling shows that researchers benefit when systems ask clarifying questions, reveal intermediate representations, and support iterative refinement rather than silently producing final scripts [95–96]. Provenance-oriented work further argues that collaborative analysis needs inspectable traces: the researcher should be able to reconstruct how a result, transformation, or visualization was produced, and where human judgment entered the loop [97]. These concerns closely parallel those in writing support, suggesting that scientific writing and scientific analysis share a deeper design commonality: both are iterative knowledge-construction activities in which hidden automation can produce epistemic debt. ¹⁸

The literature on qualitative research and data curation expands this perspective beyond quantitative coding [100–103]. Here, LLMs are used for transcript handling, coding support, memo generation, and theme exploration, but scholars emphasize tensions around interpretive validity, researcher positionality, and the risk that fluent outputs conceal analytic shortcuts [100–101, 103]. The emerging consensus is that these tools may be best positioned as assistants for exploring, organizing, and contrasting interpretations, rather than as autonomous analysts. This mirrors the writing literature, where AI is often most helpful when it externalizes options and reflections instead of replacing authorial reasoning. ¹⁹

Manuscript drafting, revision, and citation grounding

Scientific drafting and revision form the most visible part of human-AI collaboration because they directly affect the manuscript submitted to peers. Recent work spans academic writing systems for citation recommendation and introduction writing [37], citation-aware drafting [38], controllable paper revision using large-scale revision corpora [39], multi-agent writing tutors integrated with authoring environments [40], and experimental studies of abstract composition and reader judgment [45–46]. These systems differ sharply in their underlying philosophy. Some still optimize for high-quality text generation. Others deliberately place the model in a reviewer, tutor, or reviser role to preserve human authorship over final wording and argument. The latter orientation is increasingly favored in HCI and writing-support research because it better preserves voice, ownership, and accountability [19–21, 28–30, 37–40]. ²⁰

A major technical advance is the shift from generic prompting to task-specific scientific revision models and datasets. ScholarCopilot treats academic writing as citation-aware retrieval and generation [38]. XtraGPT introduces a large corpus of section-level revision instructions and responses intended to support controllable academic paper revision rather than unconstrained generation [39]. Such work matters because scientific revision is not equivalent to ordinary editing. Revisions may involve strengthening claims, clarifying novelty, aligning with reviewer expectations, restructuring related work, or tempering causal language. Models trained or prompted only for fluency can easily optimize the wrong objective, producing smoother prose without improving scientific adequacy. ²¹

The human side of drafting is equally important. Studies of co-writing with LLMs find that writers care deeply about authenticity, ownership, and the balance between assistance and self-expression [17–19, 25–29, 31–35]. In educational and reflective writing contexts, designs that encourage revision of AI suggestions, preserve writer values, or frame AI as feedback rather than direct replacement appear more compatible with learning and agency than unconstrained text generation [17–18, 20–22, 30–35]. In scientific writing, these concerns generalize from “voice” to “epistemic authorship”: who actually formulated the argument, selected the evidence, and stands behind the claims? Transparent revision

histories and interaction traces are therefore not cosmetic additions; they are mechanisms for preserving the social contract of authorship. ²²

Empirical studies of AI-assisted scientific writing complicate simplistic narratives of quality. Corpus studies detect growing stylistic influence of LLMs in academic text [41, 43–44, 137, 145]. Experimental studies of abstract writing and research summaries show that readers often struggle to distinguish human and AI-generated text, yet their judgments are shaped strongly by disclosure and beliefs about AI involvement [45–46, 127]. This suggests that evaluation should distinguish at least three questions: objective textual adequacy, actual human contribution, and perceived legitimacy. Scientific communities care about all three. An acceptable long-term design cannot optimize only the first. ²³

Review, publication, and post-publication support

Peer review is a natural but contentious site for human-AI collaboration. On one hand, the volume of submissions makes some form of AI assistance attractive [106–115]. On the other hand, poor review quality can distort scientific selection, and review systems are vulnerable to manipulation. Recent surveys of automated scholarly paper review now catalog methods for review generation, rebuttal support, meta-review, and after-review tasks [107, 110]. Systems such as ReviewAgents and ReviewEval decompose reviewing into structured steps and attempt to align generated critique with venue expectations [108–109]. ReviewFlow and related work instead focus on scaffolding human reviewers, especially novices, with contextual prompts and structured reflection [106]. This distinction between replacing reviewers and supporting reviewers mirrors the broader contrast between automation and collaboration found throughout the survey. ²⁴

The literature reveals both promise and danger. Some studies find that LLM-generated reviews can resemble human comments in presentation and broad critique [107–110]. Others document substantial risks, including susceptibility to hidden prompt injection in manuscripts, weak alignment with human judgments under adversarial conditions, and challenges in detecting AI-generated peer reviews [111–112]. There is also growing evidence that AI-generated review can feed back into drafting, potentially improving presentation and readability, but this benefit depends on calibration, critique quality, and the ability of authors to act selectively on feedback [113–115]. As with writing, the best-supported use case at present is not wholesale replacement of reviewers, but carefully bounded collaboration with explicit provenance, auditability, and human responsibility for final recommendations. ²⁵

Finally, post-publication and workflow-level benchmarking are emerging as crucial complements to review generation. Benchmarks such as CORE-Bench shift attention from fluent scholarly text to whether systems can support reproducibility and credible scientific verification [116]. This is a valuable correction. If the field evaluates research assistants only on the persuasiveness of their prose, it will systematically overestimate readiness for actual scientific deployment. Workflow support must be measured by whether it helps researchers produce reliable artifacts, not merely polished narratives. ²⁶

How the field evaluates collaboration

Evaluation remains one of the least mature parts of the area. Many papers still evaluate point capabilities such as BLEU, ROUGE, acceptance of suggestions, or user preference, which are insufficient for scientific collaboration [10, 30, 45, 65, 73–76, 88, 91–92, 109, 115–116]. The problem is structural: a system may generate locally plausible text while increasing global verification costs, obscuring provenance, or inducing overreliance. Recent evaluation frameworks therefore argue for multi-dimensional assessment that includes task success, calibration, workflow efficiency, interpretability, accountability, and human outcomes such as learning or confidence [10, 21, 31–35, 66–67, 95, 106, 109, 115]. ²⁷

A useful way to synthesize current practice is to distinguish four layers of evaluation. The first is artifact quality: textual coherence, code correctness, review constructiveness, or retrieval relevance. The second is evidential grounding: citation faithfulness, claim support, transparency of revisions, or provenance of analysis. The third is collaboration quality: user agency, ownership, trust calibration, selective reliance, and workload redistribution. The fourth is institutional fit: disclosure adequacy, compatibility with authorship norms, robustness to gaming, and contribution to reproducibility. Much of the recent literature can be read as a gradual migration from the first layer toward the latter three [21, 27–29, 45–46, 66–76, 106–127]. ²⁸

Challenges and Prospects

Reliability, provenance, and reproducibility

The foremost technical challenge is that scientific collaboration requires verifiable grounding, not only fluent generation. Reference hallucination, weak claim attribution, unstable literature retrieval, and silent analytic assumptions can all produce convincing but unreliable scientific artifacts [65, 74–76, 93–99, 111–116, 136]. The literature increasingly shows that citation faithfulness is a stronger bottleneck than surface prose quality. This is especially visible in citation benchmarks and reference-hallucination studies, where progress remains modest relative to expert human performance [74–76, 136]. Prospective systems will likely need claim-level provenance, persistent evidence objects, and interaction designs that make verification part of the normal workflow rather than an expensive afterthought. ²⁹

Reproducibility is the second major fault line. In programming, data analysis, and experiment support, the risk is not only wrong output but irrecoverable reasoning: researchers may inherit code, parameter settings, or transformation logic that they cannot fully explain [93–105, 116]. In writing, the analogous risk is laundering unsupported claims through fluent prose. One important future direction is therefore workflow-native provenance—systems that record which claims came from which papers, which analyses came from which code paths, which revisions were accepted or rejected by the human, and where uncertainty remained unresolved. Such provenance is valuable not only for trust, but also for pedagogy, auditing, and later reproduction by collaborators or reviewers. ³⁰

Human agency, expertise, and learning

A second challenge concerns the redistribution of cognitive labor. Human-AI collaboration can save time, but it can also shift work into screening, editing, and validating machine output [5–7, 17–18, 21–22, 31–35, 66–67, 93–99]. This issue is particularly acute in education and early-career research. Tools that produce polished text too easily may reduce opportunities for conceptual struggle and rhetorical development; tools that provide critique and scaffolded reflection may instead strengthen learning [20–22, 30–35, 106]. The broader productivity literature helps explain why results are mixed: generative AI often increases average task speed, but benefits are heterogeneous and can interact with expertise, task structure, and the ability to evaluate outputs [128–130, 138–139]. ³¹

This means that future systems should be designed with differentiated collaboration modes rather than a single “assistant” paradigm. Novices may need pedagogical scaffolds, explanation, and encouraged rewriting. Experts may prefer rapid search augmentation, deep citation control, or configurable agent pipelines. Team settings may need shared provenance and explicit contribution tracking. The same model output can therefore be beneficial or harmful depending on who uses it, when, and under what institutional pressure. Human-AI collaboration research in science will mature only when it measures these conditional effects directly. ³²

Integrity, disclosure, and authorship

Research integrity issues now sit at the center of the field rather than its periphery. Systematic analyses of ethical challenges across research phases identify concerns about plagiarism, bias, fabricated references, confidentiality, authorship inflation, and unequal access [117–121, 127, 130, 140]. Scientific writing raises these issues with unusual sharpness because publication confers credit and trust. As a result, the literature increasingly treats disclosure not as an administrative footnote but as a design and governance problem. Proposals such as AI-use reporting cards, cross-journal analyses of disclosure practices, and transparency tools for co-writing all point toward the need for machine-readable contribution records that can accompany manuscripts, reviews, and revisions [118–119, 127]. ³³

Authorship is especially contested. Studies of collaborative writing, authorship analysis, and disclosure perception show that readers and institutions care not only whether AI was used, but how human effort remained legible in the final artifact [122–127]. The most productive framing in the literature is not to ask whether an LLM is literally an “author,” but whether current workflows let humans retain substantive control over claims, evidence, and revisions. If not, conventional authorship categories become strained. The likely near-term solution is neither full prohibition nor casual normalization, but more granular contribution reporting aligned with human responsibility for verification and final approval. ³⁴

Infrastructure, policy, and future research agenda

The long-term prospects of the area depend on infrastructures rather than isolated models. Three infrastructural needs are already clear. The first is scholarly grounding infrastructure: open corpora, citation graphs, claim-attribution benchmarks, reproducibility benchmarks, and stable paper APIs [74–76, 116, 131–136]. The second is user-facing workflow infrastructure: integrated authoring, annotation, and provenance environments that fit existing research habits instead of forcing scholars into black-box chat interfaces [21, 39–40, 59–60, 97]. The third is institutional infrastructure: disclosure schemas, contribution taxonomies, reviewer guidance, and evaluation norms for AI-assisted submissions and reviews [106–116, 117–127]. ³⁵

A final prospect is that the field may evolve away from the language of “copilot” toward a richer theory of scientific collaboration with AI. Recent critiques correctly note that “human-AI collaboration” can become an empty slogan unless it specifies power, control, incentives, and task asymmetries [9, 142]. Scientific workflows make these issues concrete. A good collaborative system should not merely increase output volume. It should help people reason better, find stronger evidence, revise more transparently, learn more effectively, and produce scientific claims that remain accountable to human communities. On that standard, the future of the field is promising, but still far from routine autonomy. ³⁶

Conclusion

The literature on human-AI collaboration in scientific writing and research workflows now spans a wide arc: from writing-support interfaces and citation recommendation to agentic systems that attempt to automate large parts of the research lifecycle. Across this diversity, one conclusion is consistent. Scientific collaboration with AI works best when models are embedded in workflows that preserve human judgment over evidence, revision, and accountability. The highest-value systems are typically not those that maximize text substitution, but those that minimize search friction, externalize alternatives, scaffold critique, track provenance, and support selective delegation [14, 19–21, 37–40, 59–60, 66–67, 93–99, 106]. ³⁷

For the next phase of the field, the central question is no longer whether LLMs can produce scientific-looking prose. That has been answered. The harder and more important questions concern whether human-AI systems can produce attributable, evidence-grounded, reproducible, and pedagogically sound science. Answering those questions will require tighter integration between HCI, NLP, information retrieval, digital libraries, science of science, and research ethics. It will also require a shift from artifact-centric evaluation toward workflow-centric evaluation of joint performance. On those terms, human-AI collaboration in science should be viewed not as a finished technology, but as an emerging socio-technical research program. ³⁸

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